1. Maturity and Duration

Maturity & Duration

- Need a summary statistics of how long a project will last.
- Simplest measure is given by **maturity**: date of the last payment.
- Consider a bond which pays 1 unit in 30 years - this bond has a maturity of 30 years.
- Consider another bond which pays 1 unit in 1 year and 1 unit in 30 years - this bond has a maturity of 30 years as well.
- Consider a third bond which pays 100 units in 1 year and 1 unit in 30 years - this bond also has a maturity of 30 year.

duration $= \frac{1\times1 + 1\times30}{1+1} = \frac{31}{2} = 15.5$

duration $= \frac{100\times1 + 1\times30}{100+1} = \frac{130}{101} = 1.29$

- **Duration**: payment-weighted average of all "payout times", and is equal to

$$\frac{\sum_t C_t t}{\sum_t C_t}.$$

duration $= \frac{\sum_t C_t\, t}{\sum C_t}$

$C_t \rightarrow$ payment

$t \rightarrow$ time.

- For bond which pays 1 unit in 1 year and 1 unit in 30 years, duration equals

$$\frac{(1\times1)+(1\times30)}{1+1} = 15.5$$

years.

- For bond which pays 100 units in 1 year and 1 unit in 30 years

$$\frac{(100\times1)+(1\times30)}{100+1}$$

or approximately 1.29 years.

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- Another commonly used measure is **Macauley Duration**.
- Instead of weighted average of cashflows, Macauley Duration is a weighted average of present values:

$$\frac{\sum_t PV(C_t)\, t}{\sum_t PV(C_t)}.$$

Macauley Duration $= \frac{\sum PV(C_t)\cdot t}{\sum PV}$ (present value)

- Idea is to discount future payouts more, since earlier earlier payments are worth relatively more.
- Note that Macauley Duration depends on the interest rates, and hence it will change when the interest rate changes.

- Consider a project with a payment of 100 units next year and 1 unit in 30 years.
- Suppose the 1-year interest rate is 2% and the 30 year interest rate is 5%.
- The project is worth

$$\frac{100}{1.02} + \frac{1}{1.05^{30}} \approx 98.04 + 0.23 = 98.27.$$

and its Macauley Duration is

$$\frac{(98.04\times1)+(0.23\times30)}{98.27} \approx 1.07.$$

Normal duration $= \frac{100\times1 + 1\times30}{100+1} = \frac{130}{101} = 1.29$

for Macauley duration $PV(100) = \frac{100}{1.02} = 98.04$

& $PV(1) = \frac{1}{(1.05)^{30}} \simeq 0.23$

$\therefore$ M' duration $= \frac{98.04\times1 + 0.23\times30}{98.04+0.23} = 1.07$

Inflation

- Inflation is the increase in the price of goods.
- With inflation, the price level is rising, and money loses its value.
- If inflation is 10% and is equal across all goods, then an item that costs 100 units today will cost 110 units next year.

Consumer Price Index

- The first step is to survey households to determine composition of the typical household's "basket" of goods and services.
- Next step is to collect data on consumer prices (for both current and base year) of all the items in the basket.
- CPI is then calculated as

$$CPI = \frac{\text{Cost of basket in current year}}{\text{Cost of basket in base year}} \times 100$$

- Consider an economy with N goods.
- p_t^i: price of good i at time t.
- q_t^i: quantity of good i at time t.

$$CPI = \frac{\sum_i p_C^i q_B^i}{\sum_i p_B^i q_B^i} \times 100$$

$$CPI = \frac{\sum p_C^i q_B}{\sum p_B^i q_B} \times 100$$

$P_C \rightarrow$ Current price
$P_B \rightarrow$ Base Price.
$q_B \rightarrow$ quantity

Ex.

	Atta. → 10 kg,	Rice → 5 kg,	Milk → 10 ltr,	dal → 2 kg.
Base price	20 Rs/kg	30	40	80
Current price	22 Rs/kg	35	50	100

$$\therefore CPI = \frac{10\times22 + 5\times35 + 10\times50 + 2\times100}{10\times20 + 5\times30 + 10\times40 + 2\times80} = \frac{220+175+500+200}{200+150+400+160}$$

$$= \frac{1095}{910} \times 100 = 1.20 \times 100$$

$$CPI/100 = \frac{\sum_i p_C^i q_B^i}{\sum_i p_B^i q_B^i} = \frac{p_C^1 q_B^1}{\sum_i p_B^i q_B^i} + \frac{p_C^2 q_B^2}{\sum_i p_B^i q_B^i} + \dots = \frac{p_B^1 q_B^1}{\sum_i p_B^i q_B^i}\frac{p_C^1}{p_B^1} + \frac{p_B^2 q_B^2}{\sum_i p_B^i q_B^i}\frac{p_C^2}{p_B^2} + \dots = \sum_i f_i \frac{p_C^i}{p_B^i}$$

$$CPI = \frac{\sum p_C^i q_B^i}{\sum p_B^i q_B^i} \times 100 \quad \therefore \frac{CPI}{100} = \frac{p_C^1 q_B^1}{\sum p_B^i q_B^i} + \frac{p_C^2 q_B^2}{\sum p_B^i q_B^i} + \dots$$

$$\therefore \frac{CPI}{100} = \frac{p_B^1 q_B^1 \cdot p_C^1}{(\sum p_B^i q_B^i) p_B^1} + \frac{p_B^2 q_B^2 \cdot p_C^2}{(\sum p_B^i q_B^i) p_B^2} + \dots$$

$$\therefore \frac{CPI}{100} = \sum_i f_i \frac{p_C^i}{p_B^i}, \quad f_i = \frac{p_B^i q_B^i}{\sum p_B^i q_B^i}$$

↳ weighted average of change in price.

- CPI can be interpreted as the weighted averages of changes in price of each goods, where the weights are given by the relative importance of each good in (base year) GDP.

Weights.

Category	Urban (weight in 2012)	Rural (weight in 2012)
Food and Beverages	36.29	54.18
Clothing and Footwear	5.57	7.36
Housing	21.67	—
Fuel and Light	5.58	7.94
Pan, Tobacco and Intoxicants	1.36	3.26
Misc.	29.53	27.26

In Urban → Housing has significant wt, ⊘ in Rural.
In rural → food & Beverages have the highest wt

- CPI is an accurate measure of the cost of selected goods that make up the typical basket, but it may not exactly measure the cost of maintaining a given standard of life.
- Some of the important issues involved are (1) substitute bias, (2) introduction of new goods and (3) unmeasured quality changes.

GDP

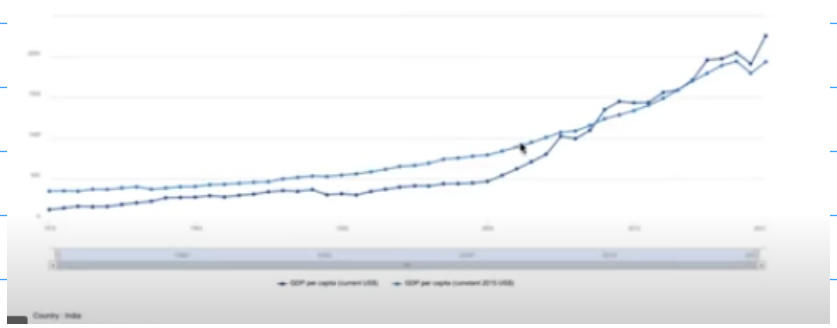
- GDP is market value of final goods and services produced in the economy during a given period.
- Market value is measured as the product of prices and quantities.
- Consider an economy which produced 50 chocolates and 20 shoes last year.
- Price of a chocolate and a shoe last year were 100 and 250 respectively.
- Nominal GDP is the value of GDP using current prices.
- Nominal GDP last year was thus
  $(50 \times 100 + 20 \times 250) = 5000 + 5000 = 10000.$

- This year the economy again produced 50 chocolates and 20 shoes.
- Price of a chocolate and a shoe this year are 200 and 500 respectively.
- Nominal GDP this year is thus
  $(50 \times 200 + 20 \times 500) = 10000 + 10000 = 20000.$
- (Nominal) GDP has doubled, even though output remains unchanged.

- Nominal GDP can change due to changes in quantites or prices.
- To focus on GDP as a measure of changes in quantities, one can use Real GDP.
- Real GDP measures the value of production using constant prices (price in some base year).
- Real GDP in the current year (using last year's prices as base): $(50 \times 100 + 20 \times 250) = 5000 + 5000 = 10000$.

∴ Real GDP remained same.

Nominal vs Real GDP.



- Another measure that is used in measuring inflation is the **GDP deflator**.
- Increases in nominal GDP can come either from an increase in real GDP, or from an increase in prices.
- If nominal GDP rises faster than real GDP, difference comes from an increase in prices.
- GDP deflator (P_t) for a particular year, is defined as the ratio of nominal GDP to real GDP for that year:

$$P_t = \frac{\text{Nominal GDP in year t}}{\text{Real GDP in year t}}$$

- **Inflation** is a sustained rise in the general level of prices—the price level.
- The **inflation rate** is the rate at which the price level increases.
- GDP deflator gives the average price of output that is the final goods produced in the economy.
- Consumers care about the average price of consumption—the goods they consume.
- These two are not the same:
  1. some of the goods in GDP are sold not to consumers
  2. consumers also purchase goods produced abroad.

- Another measure that is often discussed is the **Wholesale Price Index (WPI)**
- WPI is based on prices that are paid to wholesalers by other producers.
- The WPI basket has three broad components:
  1. primary articles weightage of 22.62,
  2. fuel and power with a weightage of 13.15,
  3. manufactured products having the biggest share at 64.23.

- Core inflation is another measure that is used to measure inflation.
- It measure the long-run trend in prices
- It typically excludes more volatile categories such as food and fuel.
- In case of India, this leads to elimination of a significant portion of the commodity basket.

Interest rates.

2 types.

- Interest rates: What one has to pay for taking a loan.
- Interest rates: Earnings one gets on savings placed in a bank.
- Interest rates: Intertemporal terms of trade - the price of shifting between periods.
- Example: suppose there are two periods t and $t+1$.
- If one has wealth of 100,000 at the beginning of period t, one can choose between spending it all in period t, or earning interest at rate i_t in period t and then spending $100,000(1+i_t)$ or some combination of the two options.

- In Q1 1980, short term interest rate in the United States was 15.1% per annum.
- In Q1 2007, short term interest rate in the United States was 5.3% per annum.
- Was it cheaper to borrow in Q1 2007 compared to in Q1 1980?
- Interest rate tells us how much money (in terms of currency) would one have to pay in the future in exchange for having one more unit of money today.
- Thus a short term interest rate of 5.3% implies that borrowing one dollar would imply needing to pay back 1.053 dollars one year later.

- When one borrows, one is actually interested in how many goods one will have to give up in the future in exchange for the goods that one gets now.
- Similarly when one lends, one is interested in how many goods one will have to gain in the future in exchange for the goods that are being given up now.
- What matters is how many goods, rather than how much money.
- The presence of inflation makes this distinction important.

Example today you borrow 100 Rs at 15% & you take 10 bananas. (banana rate = 10/unit. year.
After an year Banana rate is 12/unit. (inflation)
So, next year you can take only $\frac{100}{12}$ 8.3 bananas.

- Interest rates expressed in units of currency are called **nominal interest rates**.
- The interest rates reported in newspapers are typically nominal interest rates.
- If the nominal interest rate for year t is i_t, borrowing 1 rupee requires one to repay $1 + i_t$ next year.

Very imp. concept.

- Interest rates expressed in terms of baskets of goods are called **real interest rates**.
- If the real interest rate for year t is r_t, borrowing the equivalent of 1 basket of goods that year requires one to repay the equivalent of $1 + r_t$ baskets of good next year.
- The real interest rates are the ones that are critical to most saving and investment decisions, and to macroeconomic analysis.

- What is the relationship between the real and the nominal interest rate?
- We need to adjust the nominal interest rate to take into account expected inflation.
- Assume there is only one good in the economy.
- Denote the one year nominal interest rate by i_t and the one year real interest rate by r_t.

- Suppose the price in currency units of one unit of the good this year is P_t.
- Then one will have to borrow P_t units of currency today.
- To repay this amount of money back one year later, one will have to pay back $P_t(1 + i_t)$ units of currency next year.

Lets say $P_t = 100$, $i_t = 10\%$.
∴ you borrow 100, Purchase one unit.
now next year you may pay $P_t + P_t \cdot i_t$
$= P_t(1 + i_t) = 100 \times 1.1 = 110$.

Next year Price is 120 (for ex.)
∴ next year you can take any $\frac{110}{120}$
∴ when you pay back, you expect to forego $\frac{110}{120}$ goods

- What one cares about is the amount of goods that have to be foregone next year to repay back.
- Let P^e_{t+1} be the expected price of the good next year.
- The amount of goods one is expected to forego next year to repay back the loan is $\frac{P_t(1+i_t)}{P^e_{t+1}}$.

- Thus we get

$$1 + r_t = \frac{P_t(1 + i_t)}{P^e_{t+1}}.$$

- Noting that expected inflation equals $\pi^e_{t+1} = (P^e_{t+1} - P_t)/P_t$, the above relation can be written as:

$$1 + r_t = \frac{(1 + i_t)}{(1 + \pi^e_{t+1})}.$$

- When the interest rates and expected inflation rate are small enough, this can be approximated as

$$r_t \approx i_t - \pi^e_{t+1}.$$

$i_t \rightarrow$ nominal interest rate
$r_t \rightarrow$ real interest rate.

$\therefore 1 + r_t = \frac{P_t(1 + i_t)}{P^e_{t+1}}$ ★ Imp.

now inflation $\pi^e_{t+1} = \frac{P^e_{t+1} - P_t}{P_t}$.

$\therefore P_t(\pi^e_{t+1} + 1) = P^e_{t+1}$ Very imp ★

$\therefore 1 + r_t = \frac{1 + i_t}{1 + \pi_{t+1}}$ ✓

$\therefore 1 + r_t = \frac{1 + i_t}{1 + \pi_{t+1}}$, r_t → real interest rate
i_t → nominal interest rate
π → inflation.

Interest Rates

- Economists usually account for expectations in one of the following ways:
- Static - Inflation is not expected to change at all.
- Adaptive - Peope adapt expectations to recent historical experience.
- Rational - Expectations are consistent with their model of how the economy works.
- Data-based - Expectations are derived from surveys of economic agents or inferred from financial market data.

Interest Rates

- In Q1 1980, short term interest rate in the United States was 15.1% per annum.
- In Q1 2007, short term interest rate in the United States was 5.3% per annum.
- Was it cheaper to borrow in Q1 2007 compared to in Q1 1980?
- Using data from OECD Inflation forecast, the nominal interest rates was 15.1% per annum in Q1 1980 and 5.3% in Q1 2007.
- The expected inflation rate was 14.2% in Q1 1980 and 2.4% in Q1 2007.
- The corresponding real interest rates was approximately 0.8% and 2.8% respectively.
- It was cheaper to borrow in Q1 1980 ✓

nominal

$\therefore 1 + r_t = \frac{1 + i_t}{1 + \pi_{t+1}}$

$\therefore$ for 1980 $1 + r_t = \frac{1 + 0.151}{1 + 0.142} = \frac{1.151}{1.142} = 1.0078$

$r_t = 0.0078 \approx 0.8\%$.

for 2007, $1 + r_t = \frac{1 + 0.053}{1 + 0.024} = \frac{1.053}{1.024} = 1.0283$

$\therefore r_t = 0.0283 = 2.8\%$.

$\therefore$ real interest rate in 2007 is higher.

L3 Inflation NPV

Inflation in NPV.

Imp. points -

- When one uses NPV, should one use the real interest rate or the nominal interest rate?
- When one uses the NPV formula, always: ✓
- discount nominal cash flows with nominal interest rates,
- discount real cash flows with real interest rates.
- Do not discount nominal cash flows with real interest rates, or real cash flows with nominal interest rates.

Example

- **Example 1A**: Suppose that the inflation rate is 100% per year and that one can buy a bond that promises a nominal interest rate of 700%.
- What is the real rate of return?
- Suppose that one rupee can buy 1 basket of good today.
- An inflation rate of 100% means that prices double next year.
- So one would need two rupees to buy 1 basket of good next year.
- One rupee invested in the bond will return 8 rupees next year.
- 8 rupees next year is the equivalent of 4 baskets of good next year.

$1+r_t = \frac{1+7}{1+1} = 4 \quad \therefore r_t = 3$ i.e. 300%. (real)

- **Example 1B:** What is the value of a project that gives 12 baskets of good next year?
- Method 1: Discount real cash flows with real interest rates: present value equals

$$\frac{12}{1+300\%} = \frac{12}{4} = 3$$

baskets of good today.

- Method 2: Discount nominal cash flows with real interest rates: present value equals

$$\frac{24}{1+700\%} = \frac{24}{8} = 3$$

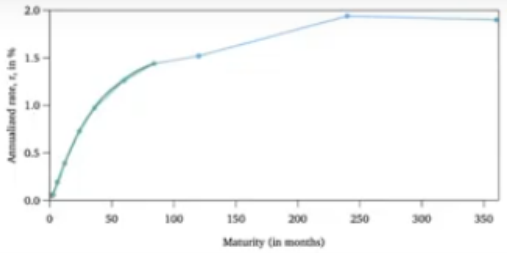
rupees today.

Real. NPV: $\frac{12}{1+3} = 3.$

Nominal inflation is 100%, ∴ 12 becomes 24 (nominal)
know we can use nominal rates.
$\frac{24}{1+7} = 3$
by both methods, we get same.

Yield Curve.

- Long term bonds can offer different interest rates than short term bonds.
- Annualized rates of return provides a way to compare rates of return over different time horizons.
- The **yield curve** shows the annualized interest rate as a function of bond maturity.
- The yield curve is also referred to as the **term structure of interest rates** or the **term structure**.
- It is a graphical representation with the time to maturity ox the X-axis and the annualized interest rates on the Y-axis.



Months	1	2	3	6
YTM in %	0.06	0.05	0.06	0.19

Years	1	2	3	5	7	10	20	30
YTM in %	0.39	0.73	0.97	1.26	1.44	1.52	1.94	1.90

$100(1+0.06)^{1/4}$ =

$100(1+1.90)^{30}$ =

- The above yield curve is for U.S. government bonds on December 31, 2021.
- Qn: Suppose one buys a 3 month bond. How much would a 100 unit of investment turn into after 3 months?
- Ans: $100\,(1+0.06\%)^{1/4} \approx 100.015.$
- Qn: Suppose one buys a 30 year bond. How much would a 100 unit of investment turn into after 30 months?
- Ans: $100\,(1+1.90\%)^{30} \approx 175.884.$

- **Note**: In the exercises below for this yield curve, we assume that we are discussing zero bonds, that is bonds without interim coupon payments.
- **Qn**: Suppose one invests 100,000 units in two-year bonds on December 31, 2021. What would this return on December 31, 2023?
- The yield curve provides information about the two year annualized rate, r_2, from this need to figure out the two-year holding rate $r_{0,2}$.
- The relation is given by

$$r_{0,2} = (1+r_2)(1+r_2) - 1 = (1.0073)\,(1.0073) - 1 \approx 1.465\%.$$

- Ans: The 100,000 units will turn into

$$100,000\,(1+1.465\%) \approx 101465$$

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$= 100000\,(1+0.0073)^2 = 101465$

- **Qn**: What if one invests 100,000 units in thirty-year bonds?
- The 30-year holding rate of return return is given by

$$r_{0,30} = 1.0190^{30} - 1 \approx 1.759 - 1 \approx 75.9\%.$$

- Ans: The 100,000 units will turn into

$$100,000\,(1+75.9\%) \approx 175884.$$

$= 100000\,(1+0.019)^{30}$

$= 175883.67$

- Qn: What is the yield for a 4-year bond?
- Though this information is not directly listed in the table above, one can interpolate annualized interest rates on the yield curve.
- Ans: The 4-year bond would have an annualized yield of $(0.97\% + 1.26\%)/2 \approx 1.12\%.$ ✓
  3yr 5yr.
- Note that this is not exact, since the yield curve is not linear.

- **Qn**: Based on the yield curve, what's the 1-year interest rate from December 2022 to December 2023?
- This would be the interest rate $r_{1,2}$: interest rate that begins in one year and ends in two years.
- This is an example of forward interest rate, or forward rate. ?

$$r_{1,2} = \frac{(1+r_2)^2}{1+r_1} - 1$$

$$= \frac{(1+0.0073)^2}{1+0.0039} - 1$$

$$= 0.0107 = 1.07\%$$

- The 1-year annualized rate is $r_1 = 0.39\%$, which is also the holding rate of return, $r_{0,1}$.
- The 2-year annualized rate is $r_2 = 0.73\%$.
- This gives the two-year holding rate of return as

$$r_{0,2} = (1+r_2)^2 - 1 \approx 1.465\%.$$

- We know that

$$1 + r_{0,2} = (1 + r_{0,1})(1 + r_{1,2}).$$

- Ans: The forward rate $r_{1,2}$ equals 1.07%.
- One can similarly calculate other forward rates such as $r_{2,3}$ and so on.

The key differences between holding rates of return and annualized rates of return lie in their calculation and interpretation.

Holding Rates of Return (r0,t)

Characteristics

- Represents the total cumulative return over the entire investment period
- Does not account for the time length of investment
- Expressed as a single percentage for the whole period
- Calculated as: $r_{0,t} = \frac{Final\ Value - Initial\ Value}{Initial\ Value}$

Example

If an investment grows from \$100 to \$121 over two years:

- Holding return = 21%
- This is the total return regardless of how long it took to achieve

Annualized Rates of Return ($\bar{r}t$)

Characteristics

- Converts the holding period return into a yearly equivalent rate
- Accounts for the time value of money and compounding
- Allows comparison of investments with different time horizons
- Calculated as: $\bar{r}t = (1 + r_{0,t})^{1/t} - 1$

Example

For the same investment above:

- Annualized return = $(1 + 0.21)^{1/2} - 1 = 10\%$
- This means the investment earned an equivalent of 10% per year

Investment Horizon and Bond Payoffs

- Suppose that you have an option between (1) buying a three-year zero-coupon bond which gives an annualized rate of 0.97% and (2) buying a 1-year zero-coupon bond which gives an annualized rate of 0.39%.
- You know that you need money one year later.
- Can you buy the long-term bond?

Yes, you can purchase a 3-yr bond & sell it after one year.

- Can you buy the long-term bond?
- Yes, since you can sell it one year later.
- However, in a perfect and certain market, you will get a return of 0.39% using this investment strategy.

YTM → first year 0.39%, second year 0.73

$1+r = (1+0.0073)^2 = 1.00146 \therefore r = 1.46\%$

- Suppose each of the bond pays off 100 units.
- Price of three-year bond today will be equal

$$\frac{100}{1.0097^3} \approx 97.15.$$ ✓

- When you try to sell it next year, it will be a two-year bond with an interest rate of 1.07% in the first year, and 1.46% in the second year.
- This follows from solving for $r_{2,3}$ from

$$(1.0097)^3 = (1.0039)(1.0107)(1 + r_{2,3})$$

Months	1	2	3	6
YTM in %	0.06	0.05	0.06	0.19

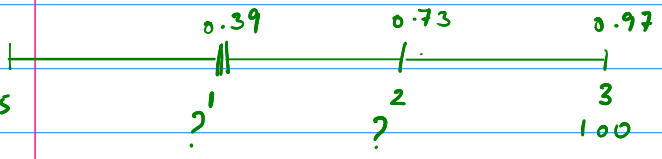
Years	1	2	3	5	7	10	20	30
YTM in %	0.39	0.73	0.97	1.26	1.44	1.52	1.94	1.90

Suppose you buy 3-year bond (YTM: 0.97) & want to sell next year.

PV: $\frac{100}{(1.0097)^3}$ (for bond of value 100) : 97.15

1 yr YTM : 0.39, 2 yr YTM : 0.73, 3 yr YTM : 0.97

If you sell the bond after 1 year, what price will you get.



yr 97.15 ? ? 100

forward rate$_1$: $\frac{(1.0073)^2}{1.0039} - 1$: 0.0107

forward rate$_2$: $(1.0073)^2 - 1$: 0.0146

After one year value : 97.15 + 0.0039 × 97.15 : 97.52.

& PV : $\frac{100}{1.0107 \times 1.0146}$: 97.52 ∴ both are same.

- Are long-term bonds riskier than short-term bonds?
- Consider the case of safe government bonds - no default risk.
- Different kind of risk due to changes in interest rate.
- What are the effects of sudden interest rate changes before maturity of bond values?
- Equal-sized interest rate movements have more impact on long-term bonds than short-term bonds.

Interest Rate Changes

- Consider a 30-year zero-bond which promises to pay 1000 units after 30 years.
- Suppose the annualized interest rate is 1.90%.
- The bond would cost $1000/1.0190^{30} \approx 568.56$ currently. ✓
- Suppose that the prevailing interest rate increases by 10 basis points to 2.0%.
- Bond value decreases to $1000/1.02^{30} \approx 552.08$ currently.
- Supposed instead the prevailing interest rate decreased by 10 basis points to 1.8%.
- Bond value would increase to $1000/1.0180^{30} \approx 585.55$ currently.

PV_1 : $\frac{1000}{(1.019)^{30}} \approx 568.56$.

PV_2 : $\frac{1000}{(1.02)^{30}} \approx 552.08$.

PV_3 : $\frac{1000}{(1.018)^{30}} \approx 585.55$

- Effect of a 10-basis-point change in the prevailing 30-year annualized yield induces an immedatiate percentage change (an instant rate of return) in the value V of the bond of

if interest rate ↑

$$r = \frac{552.08 - 568.56}{568.56} \approx -2.9\%$$ ✓

for an up 10 bps movement, and a change in the value V of the bond of

↓

$$r = \frac{585.55 - 568.56}{568.56} \approx +3\%$$

for a down 10 bps movement.

- Every 10 lakh invested in 30-year bonds exposed one to about 30,000 in instant risk for every 10-basis-point yield change in the economy.

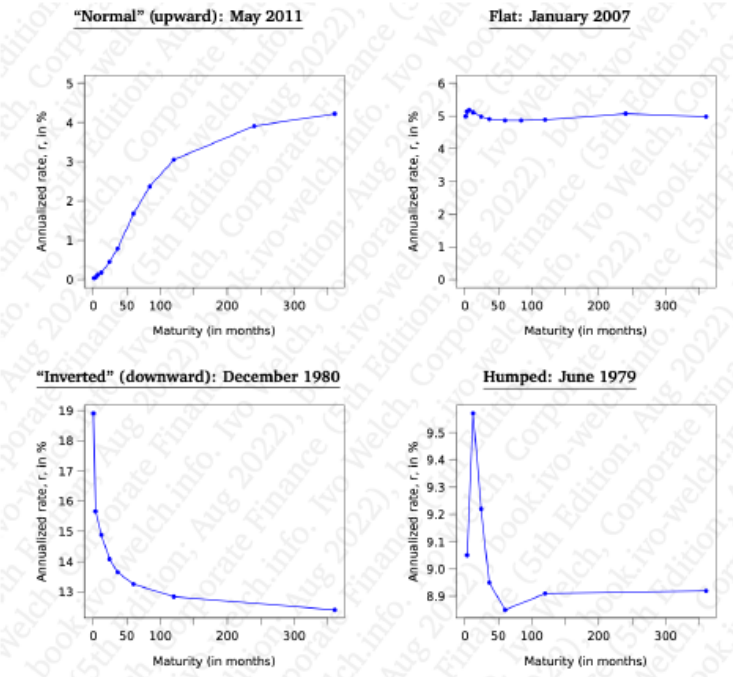
- Suppose instead we consider a 1-year bond.
- Suppose that the bond has an interest rate of 1.90%.
- Consider the same 10-basis point change in the prevailing interest rate.
- Value of 1-year bonds are 981.35 at 1.90%, 982.32 at 1.80% and 980.39 at 2.00%.

∴ in short term - not so much of difference.

- An equal-sized change in the prevailing interest rate affects long-term bonds more strongly.
- If the bond is due soon, changes in interest rate have much less influence.
- One can get one's money back and then decide whether to consume or reinvest at the new rate.
- A long-term bond in contrast can lose or gain a lot of value.
- Only the most short-term bond can be considered risk-free.

Term-Varying Costs of Capital

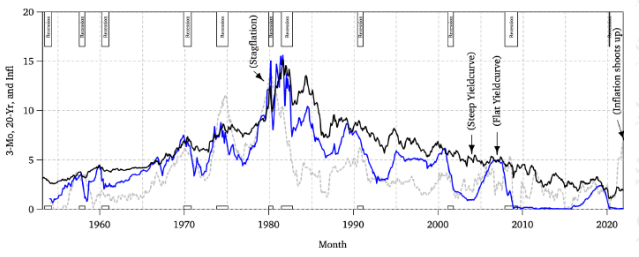
- The appropriate cost of capital should usually depend on how long-term the project is.
- The term structure is usually upward-sloping: short term projects usually have lower costs of capital.
- Longer term projects, even if they are not more likely to default, often face a higher cost of capital, and thus should have to deliver higher returns.
- The difference between long-term and short-term rates is called the **term premium**, or the time premimum.



- Yield curves are often classified into the following cases:

1. **Flat**: Little to no difference between short-term and long-term rates.
2. **Upward-sloping**: Short term rates are lower than long-term rates.
3. **Downward-sloping**: Short-term rates exceed long-term rates.
4. **Humped**: Short-term rates and long-term rates exceed medium-term rates.

Yield Curves



Blue line: yield on 3-month United States Treasury bills
Black line: yield on 20-year United States Treasury bonds

- Perception of yield curve as a useful, though unreliable and noisy signal, of where the economy is heading.
- Steep yield curves often follow emergence from a recession.
- Inverted yield curves sometimes signal an impending recession.
- Yield curves are determined by the demand and supply for savings and credit in the economy.
- Central banks can influence but not control it.

- The upward slope is the most common shape.
- Let's consider a two-year case.
- Suppose the yield curve indicates that the 1-year rate is $r_1 = 5\%$ and the two-year rate is $r_2 = 10\%$
- The corresponding 1-year forward rate is thus $r_{1,2} \approx 15.24\%$.
- There are two possible explanations for this case: higher future interest rates and/or compensation for risk.

$$\frac{(1.1)^2}{1.05} - 1$$

$$= \frac{1.21}{1.05} - 1 = .1524 = 15.24\%.$$

- One explanation is that the 1-year interest rate next year will be higher than the current 5%.
- Perhaps next year's 1-year interest rate will be the 15.24% that one predicts it would be in a perfect world with perfect certainty.
- In equation form, this corresponds to asking if it is the case that

$$(1 + r_{1,2})\,(1 + E(r_{2,3})) = (1 + r_{1,3})$$

where $E(.)$ refers to the expected value.

- Alternatively, investors can earn higher rates of return holding long-term bonds than they do holding short-term bonds.
- If the yield curve were to be the same shape next year, then 100 units of investment in two consecutive 1-year bonds would lead to an interest of 10.25 units.
- 100 units of investment in a two-bear bond would lead to an interest on 21 units.

- In equation form, this corresponds to asking if it is the case that

$$(1 + r_{1,2})\,(1 + E(r_{2,3})) < (1 + r_{1,3})$$

- The two cases thus correspond to asking whether current higher long-term interest rate predicts higher interest rates in the future, or whether they offer extra compensation for investors willing to hold long-term bonds.

- When inflation is high, investors typically demand higher interest rates.
- One potential explanation for upward sloping yield curve is that investors believe that inflation would be higher in the future.
- Since inflation erodes the value of higher interest rates, interest rates should be higher to compensate for the lesser value of money.
- Note that this argument applies only to a yield curve corresponding to a financial security that pays off in nominal terms.
- It should not apply to any bond payoffs that is inflation-indexed.

- There are inflation-indexed bonds as well, such as Treasury Inflation Protected Securities (TIPS).
- These contracts for instance pay out promised interest rate plus the (CPI) inflation rate.
- These bonds should not be affected by inflation in a perfect market.
- If the nominal yield-curve is upward sloping just because of higher future inflation rates, then a yield curve based on bonds which pay in real terms should not be upward-sloping.

- Instance of annualized TIPS rate (at the end of 2021):

Maturity	5-year	7-year	10-year	20-year	30-year
TIPS Interest Rate	-1.61%	-1.31%	-1.04%	-0.63%	-0.44%

- Comparing short-term vs long-term rates on both TIPS and nominal yields shows that TIPS slope was steeper.
- Hard to say whether inflation expectation could account for entire nominal yield-curve slope.
- In most of history, they couldn't - but sometimes they can.

- Short term interests can be higher in the future because of other reasons as well.
- The 30-year yield of 1.90% could be higher than the 1-year yield of 0.39% because investors might have expected future 1-year interests to be even higher.
- Perhaps they expect capital to be more scarce in the future and investment opportunities to be better.
- Empirical evidence suggests though a high forward interest rate does not predict a high future interest rate.

- Empirical data suggests that one would have ended up historically with more money if one had bought 30-year bonds than if one had bought and rolled one-month bonds every month for 30-years.
- Why are long term bonds better investment than short-term bonds?
- Several possible explanations.

- One theory is that one might have different investor habitat and preferences.
- Different types of investors prefer different bonds, so that the markets are segmented.
- Not clear why long-term bond investors would be more scarce so that they would require more compensation, while short-term bond investors would be so abundant that the bonds can be sold at a higher price.
- Why would borrowers not just sell short-term bonds?

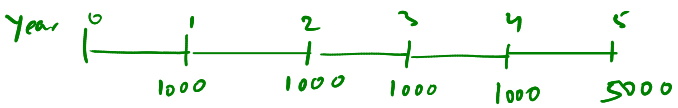
- Other potential explanations include irrationality of short-term investors, imperfect markets (taxes and liquidity).
- These explanations are unlikely to be true.
- Empirical evidence suggests the more likely explanation is that long-term rates tend to be higher as a compensation for risk.

- Interim changes in interest rates have more impact on long-term bonds.
- Rolling over short-term bonds insulates one from the risk that interest rates will change in the future.
- If one holds a one-month bond, and interest rates double by next month, one can purchase more bonds next month.
- If one instead holds a long-term bond, one would be subject to higher losses because of interest rate changes.
- Long-term bonds, being riskier, need to offer investors more return on average than short-term bonds.

GAs

1) A bond pays a coupon payment of Rs 1,000 each year, starting next year. The bond matures after 5 years and pays Rs 5,000 on maturity. If the prevailing interest rate is 8% per year, then what is the duration of the bond?

- ○ 5 years
- ✓ 4 years
- ○ 2.5 years
- ○ 1.5 years



$$\therefore \text{duration} = \frac{1000\times1 + 1000\times2 + 1000\times3 + 1000\times4 + 5000\times5 + 1000\times5}{1000 + 1000 + 1000 + 1000 + 5000 + 1000}$$

$$= \frac{40000}{10000} = 4 \text{ years.}$$

2) A bond pays a coupon payment of Rs 1,000 each year, starting next year. The bond matures after 5 years and pays Rs 5,000 on maturity. If the prevailing interest rate is 8% per year, then what is the Macauley duration of the bond?

$= \sum PV(C_t)\cdot t / \sum PV.$

- ○ 4.34 years
- ✓ 3.84 years
- ○ 3.05 years
- ○ 2.62 years

$PV(1000)_1 = \frac{1000}{1.08}$, $PV(1000)_2 = \frac{1000}{1.08^2}$, $PV_3 = \frac{1000}{1.08^3}$, $PV_4 = \frac{1000}{1.08^4}$, $PV_5 = \frac{1000}{1.08^5}$, $PV_6 = \frac{5000}{1.08^5}$

$$\therefore MD = \frac{925.92 + 857.33\times2 + 793.83\times3 + 735.14\times4 + 4083.5\times5}{925.92 + 857.3 + 793.83 + 735 + 4083.5} = \frac{28379.16}{7395.62}$$

$$= 3.84$$

3) A typical consumer basket in an economy consists of two food packets and one pair of clothes. In 2021, the price of a food packet was 30 units and the price of a pair of clothes was 150 units. In 2022, the price of a food packet was 35 units and the price of a pair of clothes was 130 units. In 2023, the price of a food packet was 40 units and the price of a pair of clothes was 170 units. Consider 2021 as the base year, so that the CPI in 2021 in normalized to 100.What is the value of the consumer price index (CPI) in 2023? **5 points**

$CPI = \frac{\sum P_c^i \cdot q_b}{\sum P_b^i q_b} \times 100$

- 95.24
- 111.85
- ✓ 119.05
- 125.00

$= \frac{80+170}{60+150} \times 100 = \frac{250}{210} \times 100$

$= \frac{2500}{21} = 119.048 \approx 119.05$

4) A typical consumer basket in an economy consists of two food packets and one pair of clothes. In 2021, the price of a food packet was 30 units and the price of a pair of clothes was 150 units. In 2022, the price of a food packet was 35 units and the price of a pair of clothes was 130 units. In 2023, the price of a food packet was 40 units and the price of a pair of clothes was 170 units. Consider 2021 as the base year, so that the CPI in 2021 is normalized to 100. What is the value of inflation rate in 2022 and 2023, respectively? **5 points**

- 4.76% and 11.85%
- -4.76% and 19.05%
- 4.76% and 19.05%
- ✓ -4.76% and 25%

In 2021 → total cost = $30 \times 2 + 150 = 210$

2022 → " = $35 \times 2 + 130 = 200$

2023 → " = $40 \times 2 + 170 = 250$

Inflation in 2022 = $\frac{200-210}{210} = \frac{-10}{210} \times 100 = -4.76\%$

Inflation in 2023 = $\frac{250-200}{200} \times 100 = \frac{50}{200} \times 100 = 25\%$.

5) An economy produces apples and buckets only. In 2020, the price of apples is 120 units per kg and the price of buckets is 60 units per bucket. In 2021, the price of apples dropped to 100 units per kg while the price of buckets increased to 100 units per bucket. In 2020, the total production of apples was 15,000 kg and a total of 100,000 buckets were produced in the economy. In 2021, the total production of apples was 20,000 kg and a total of 80,000 buckets were produced. If 2020 was the base year, then the real GDP in the year 2021 was? **5 points**

- 72,00,000
- 78,00,000
- 85,00,000
- 100,00,000

Real GDP (Keep Price Constant)

∴ real GDP of 2021 = $120 \times 20000 + 60 \times 80000$

$= 2400000 + 4800000$

$= 7200000$ ✓

6) An economy produces apples and buckets only. In 2020, the price of apple is 120 units per kg and the price of buckets is 60 units per bucket. In 2021, the price of apples dropped to 100 units per kg while the price of buckets increased to 100 units per bucket. In 2020, the total production of apples was 15000 kg and a total of 100000 buckets were produced in the economy. In 2021, the total production of apples was 20000 kg and a total of 80000 buckets were produced. If 2020 was the base year, the value of the GDP deflator in 2021 was? **5 points**

- 0.72
- 1.00
- 1.28
- ✓ 1.39

GDP deflator = $\frac{\text{Nominal GDP}}{\text{Real GDP}} = \frac{100 \times 20000 + 100 \times 80000}{7200000} = \frac{1000000}{7200000} = \frac{100}{72}$

$\frac{100}{72} = \frac{25}{18} = 1.39$

7) The average price level in the economy is 100 units today. The average price level is expected to rise to 130 units in the next period. The nominal interest rate is 15% today, then what is the value of the real interest rate today? (Note: Use the exact relation, and not the approximation.) **5 po**

- 45.00%
- -15.00%
- 13.04%
- -11.54%

$1+r_t = \frac{1+i_t}{1+\pi_t}$, $\pi_t = \frac{130-100}{100} = \frac{30}{100} = 0.3$.

$\therefore 1+r_t = \frac{1+0.15}{1+0.3} = \frac{1.15}{1.3}$ $\therefore r_t = \frac{1.15}{1.3} - 1 = -0.1154$

$= -11.54\%$

8) Which of the following is correct? is for GDP

- The GDP deflator is better than the CPI at reflecting the goods and services bought by consumers. No.
- ✓ The CPI is better than the GDP deflator at reflecting the goods and services bought by consumers. Yes.
- The GDP deflator and the CPI are equally good at reflecting the goods and services bought by consumers. No
- The core inflation only includes the volatile categories such as food and fuel. No.

9) There is a project which takes Rs 40000 as investment today. It promises to pay 1 gram of gold each year, beginning from next year, for 5 years. At the end of 5 years, the project also pays Rs 10000. The price of gold is Rs 80000 per 10 grams today. The price of gold rises by the inflation rate in the economy, which is 4% per year. If the prevailing rate of interest is 10%, then **5 point**

- An investor should not invest in the project as the payout is in the form of gold.
- An investor should not invest in the project as the net present value is negative.
- An investor should invest in the project as the net present value is positive.
- An investor is unable to decide whether to invest or not as the net present value cannot be calculated.

Lets analyse $PV = -40000 + \frac{8000 \times 1.04}{1+0.1} + \frac{8000 \times 1.04^2}{(1.1)^2} + \frac{800 \times 1.04^3}{(1.1)^3} + \frac{8000 \times 1.04^4}{(1.1)^4} + \frac{8000 \times 1.04^5}{(1.1)^5}$

$\therefore PV = -40000 + \frac{10000}{1.15} + 8000\left[\frac{1.04}{1.1} + \frac{1.04^2}{1.1^2} + \frac{1.04^3}{1.1^3} + \frac{1.04^4}{1.1^4} + \frac{1.04^5}{1.1^5} + \frac{10000}{(1.1)^5}\right]$

$= -40000 + 6209.2 + 8000\left[\left(\frac{1.04}{1.1}\right) + \left(\frac{1.04}{1.1}\right)^2 \cdots \left(\frac{1.04}{1.1}\right)^5\right]$

$= -33790.8 + 8000(A)\left[0.9454 + 0.9454^2 + 0.9454^3 + 0.9454^4 + 0.9454^5\right]$

$0.9454A = 0.9454^2 + \cdots + 0.9454^6$

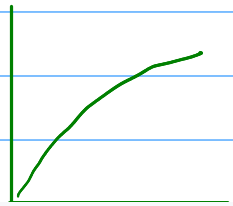
$\therefore A(1-0.9454) = 0.9454 - 0.9454^6$

$\therefore A = \frac{0.9454 - 0.9454^6}{0.0546}$

$\therefore NPV = -33790.8 + 8000 \times 4.23 = -33790.8 + 33840 = +ive$

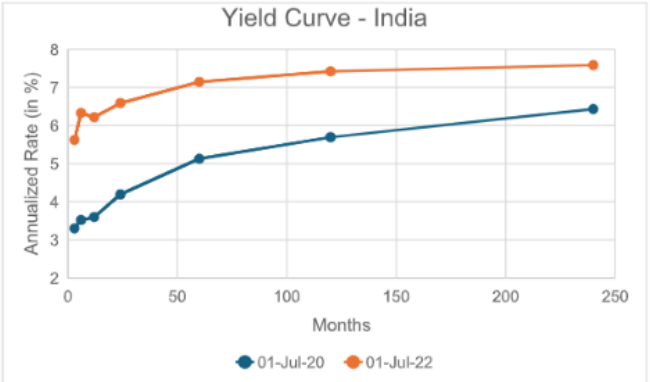
$\therefore$ Should invest.

10) An upward sloping yield curve suggests:



- Short-term interest rates are higher than long-term interest rates.
- Investors expect higher interest rates in the future.
- ✓ Long-term interest rates are higher than short-term interest rates.
- The bond market is expecting a recession.

The chart below shows the yield curve for India at two different time periods: 01-July-2020 and 01-July-2022. The interest rate values for different maturities are listed in the table below as well.



Months	01-Jul-20	01-Jul-22
3-month	3.30	5.62
6-month	3.52	6.33
1-year	3.60	6.22
2-year	4.19	6.59
5-year	5.13	7.14
10-year	5.69	7.42
20-year	6.43	7.59

11) Suppose an investor buys 6-month bonds of worth Rs 1,000 on 01-Jul-20. How much would be the value of the investment after 6-months?

- ○ Rs 965.99
- ✓ Rs 1017.45
- ○ Rs 1035.20
- ○ Rs 1071.64

$= 1000(1+r)^t = 1000(1+0.0352)^{1/2} = 1017.45$

12) Assuming the yield curve is linear between the given points. What is the annualized rate of return of a 7-year bond on 01-Jul-20?

- ○ 5.13%
- ○ 5.25%
- ✓ 5.35%
- ○ 5.59%

$= r_5 + \frac{r_{10} - r_5}{5} \times 2 = 5.13 + \frac{5.69 - 5.13}{5} \times 2$

$= 5.13 + \frac{0.56}{5} \times 2 = 5.354$

13) What is the holding rate of return of a 10-year bond purchased on 01-Jul-20?

- ○ 5.69%
- ○ 56.90%
- ✓ 73.92%
- ○ 173.92%

$1 + r = (1 + r_{10})^{10} = (1.0569)^{10} = 1.7392$

$\therefore r = 0.739157 \approx 73.92\%$

14) What is the difference between the price of 1-year bond purchased on 01-Jul-20 and 1-year bond purchased on 01-Jul-22? The face value of both the bonds is Rs 1,000 and both are zero coupon bonds.

- ○ Rs 26.20
- ✓ Rs 23.81
- ○ Rs 19.75
- ○ Rs 15.42

price of 1 yr bond purchased on 01 Jul 20 $= \frac{1000}{1.036} = 965.25$

price of 1 yr bond purchased on 01 Jul 22 $= \frac{1000}{1.0622} = 941.44$

$\therefore$ difference $= 965.25 - 941.44 = 23.81$

15) What is the forward 1-year interest rate for a bond purchase on 01-Jul-21 considering the yield curve of 01-Jul-20?

- ✓ 4.78%
- ○ 3.41%
- ○ 2.46%
- ○ 0.57%

$r_2 \rightarrow$ 2 year yield

$r_1 \rightarrow$ 1 year yield

forward rate $r_{1,2} = \frac{(1+r_2)^t}{1+r_1} - 1$

$= \frac{(1+0.0419)^2}{1+0.036} - 1 = \frac{(1.0419)^2}{1.036} - 1 = .0478$

$= 4.78\%$

16) You purchase a 2-year bond on 01-Jul-22. You intend to sell it after one year. What would be the 1-year holding rate of return of your purchase? (Assume perfect and certain market condition holds)

- ✓ 6.96%
- ○ 6.22%
- ○ 5.41%
- ○ 2.47%

To calculate the **1-year holding rate of return** for a **2-year bond purchased on 01-Jul-22** and sold after **1 year (on 01-Jul-23)** under **perfect and certain market conditions**, we use the **holding period return (HPR) formula:**

$$HPR = \frac{P_1 + C - P_0}{P_0}$$

Where:

- P_0 = **Price of the 2-year bond on 01-Jul-22** (determined using the yield curve for 01-Jul-22).
- P_1 = **Price of the remaining 1-year bond on 01-Jul-23** (determined using the yield curve for 01-Jul-23, which we assume to be the same as the forward rate for 1-year).
- C = **Coupon received after 1 year** (assuming annual coupon payments).

Lets say we purchased a bond of 1000 for 2 years.

$\therefore PV = \frac{1000}{(1.0659)^2} = 880.17$

In perfect market, value after one year will be $880.17 + 0.0622 \times 880.17$.

$\therefore$ holding rate $\rightarrow 6.22$.

After One year it becomes a One year bond & Value $= \frac{1000}{1.0622} = 941.4$

$\therefore$ holding rate $= \frac{941.4 - 880.17}{880.17} \times 100 = 6.96$

17) Consider the yield curve on 01-Jul-2022. If the prevailing interest rate increases by 20 basis points, then the value of 10-year bond

- ○ Increases by 20%.
- ○ Increases by 1.90%
- ✓ Decreases by 1.84%
- ○ Decreases by 14.82%

10 yr rate $= 7.42$

$\therefore$ Value $= \frac{1000}{1.0742^{10}} = 488.18$

Now lets say interest rate increases by 20 basis point

$\therefore$ Interest $= 7.62$

$\therefore$ Value $= \frac{1000}{1.0762^{10}} = 479.8$

$\therefore \% = \frac{479.8 - 488.18}{488.18} = -0.0184$

$\therefore \downarrow$ by 1.84%.

18) What is the 10-year to 90-days term premium on 01-Jul-22?

- ✓ 1.80%
- ○ 1.20%
- ○ 0.99%
- ○ 0.32%

• The difference between long-term and short-term rates is called the **term premium**, or the time premimum.

10 year $= 7.42$

90 days $= 5.62$

$\therefore$ prem $= 7.42 - 5.62 = 1.80$

19) Statement A: The yield curve may be downward sloping which is often called inverted yield curve. Correct.

Statement B: Due to economic uncertainty, investors sometimes find safety in long- term bonds. ✓ Correct.

- ✓ Both statement A and statement B are correct.
- ○ Both statement A and statement B are incorrect.
- ○ Statement A is correct, and statement B is incorrect.
- ○ Statement A is incorrect, and statement B is correct.

Which of the following statements is/are correct?

A yield curve is always upward-sloping. NO

A yield curve is always downward sloping. No.

Inflation does not have any impact on the yield curve. No.

Investors expect higher annualized rate of return on long-term bonds as compensation for more risk. ✓ Yes.